Evaluating Augmentation Reliability in Limited-Data Chest X-Ray Pneumonia Detection Using Grad-CAM

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This project addresses the problem of automatic pneumonia detection from chest X-ray images. The task is a binary medical image classification problem: given a chest X-ray, the model will classify it as either Normal or Pneumonia. Instead of focusing only on whether data augmentation improves classification accuracy, this project investigates a broader question: how do training-data size and augmentation strength affect both model performance and the stability of model explanations?

The motivation comes from the fact that deep learning models usually require large labeled datasets, while in medical imaging, collecting and labeling data is expensive and often requires expert knowledge. In practical medical AI scenarios, models may need to be trained on limited local data, and high accuracy alone may not be sufficient if the model relies on unstable or irrelevant visual patterns.

The project will use the publicly available Kaggle Chest X-Ray Images (Pneumonia) dataset. The dataset contains labeled X-ray images from two classes, Normal and Pneumonia, and is already divided into training, validation, and test sets. No manual data collection or annotation will be required.

The proposed method is based on transfer learning using PyTorch. We will fine-tune MobileNetV3-Small, ResNet18, and EfficientNet-B0. For each architecture and training-data size, fine-tuning without data augmentation will serve as the baseline. Comparing architectures with different model sizes will allow us to examine whether augmentation affects lightweight and standard CNN models differently under limited-data conditions.

The experimental design will compare three training-data sizes: 20%, 50%, and 100% of the training set. For each data size, we will test three augmentation regimes: no augmentation, mild medically realistic augmentation, and stronger generic augmentation. Performance will be evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

In addition to classification metrics, we will use Grad-CAM as an explainability-based auditing tool. On a fixed class-balanced subset of test images, explanation stability will be quantified using the mean cosine similarity between normalized Grad-CAM heatmaps before and after small input perturbations. We will also compare the heatmaps across different data-size and augmentation settings. The goal is not to clinically validate the explanations, but to analyze whether augmentation leads to more consistent visual evidence.

Prior work by Kermany et al. (2018) demonstrated deep-learning-based pediatric pneumonia classification from chest X-rays, while Selvaraju et al. (2017) introduced Grad-CAM for class-

discriminative visual explanations. Building on these approaches, our planned contribution is an empirical evaluation of how augmentation strength and training-data availability affect both classification performance and explanation stability across models of different sizes.

The expected outcome is a structured comparison showing when augmentation improves performance, whether stronger augmentation can harm explanation stability, and whether lightweight models behave differently from larger models under limited-data conditions.